SILTHAUL · DESIGN DECISION RECORD

SiltHaul design for construction

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DOCUMENT CONTROL

Document ID	SLH-DDR-002	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/silthaul		
Source	docs/decisions/0002-design-for-construction.md @ 9347c2b		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Constructability review and changes that make the design buildable, decided under Amish's pre-approval of 2026-10-03

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** accepted

Context

STANDARDS section 18 asks for every part to be makeable by its stated process and to fit and fasten to its neighbours (Amish, 2026-09-30: "fix the design assumptions to match and be physically feasible"). The TRL 2 concept (SLH-PRC-001 v0.2, SLH-DDR-001) was reviewed part by part in `cad/src/model.py` : how each part is made, how it joins each neighbour, and build123d checks for overlapping parts and for parts that do not touch what holds them. The checks now report no overlaps and no floating parts. Amish pre-approved every recommendation on 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." No change alters what SiltHaul does or its pitch; the safety-related changes all take the conservative side.

Options considered

For each problem the simplest physically sound fix was chosen; the alternatives are noted in Table 1.

Decision

Table 1. Changes made for construction.

#	Part	Problem found	Change	Why
1	Capstan frame	Over 25 kg in 40 x 40 x 3 tube with solid braces	40 x 40 x 2 tube throughout, 8 mm bearing plates: 24.7 kg	R9
2	Capstan frame	Sling eye at ground level let the capstan tip forward at the overload limit	Anchor bar 40 x 40 x 3 between the rear braces with an eye 200 mm up, level with the drum axis	The pulls line up; the frame is held back by 315 N m
3	Bearing pads	A 60 mm pad clashed with the first ratchet wheel	Pads 50 mm wide, 180 long	Clearance
4	Drum	Drum with its bearings was 26 kg, over R9	4 mm rings and end discs with six lightening holes; 6 mm ratchet wheels: 23.1 kg with bearings	R9
5	Ratchet and pawls	Both pawls on one pivot could not each clear the tooth behind the one they hold	Wheel 1 and its pawl mirrored to the front, on a bracket on the front brace; wheel 2 and its pawl on the post	Each pawl's body clears the teeth; nose 1.3 mm off the face
6	Chain drive	Sprocket hub inside the chain line; guard wall in the chain's plane	Hub 36 mm; guard moved out to 44 to 70 mm beyond the bearing plane; cranks moved out to suit	Clearance
7	Shear pin	Not detailed	4 mm S235 pin through a hub that turns freely on the crank shaft and the shaft itself	R11
8	Stakes	Stakes hung free in their tubes	A 32 mm head welded on rests on the stake tube	Each part rests on what holds it
9	Tail block	One large sheave would have been 300 mm; 20 mm pins at 270 MPa at the limit	Two 125 mm sheaves 175 mm apart on 25 mm pins (138 MPa), spacers, a keeper bar and four anchors	Bought sheaves; strength
10	Scraper box	Return leg could not pass the box in a 0.7 m door	Box 450 mm wide; return leg 300 mm off the box line	R4
11	Scraper box	Vertical back would bulldoze mud on the return	Back slopes 80 mm; skids cut at 45 degrees at the rear	Rides over mud
12	Bridles	Shackle and lug geometry not defined	19 mm lug holes, shackle pins through them, chain legs to pear rings 300 mm out	Fixings
13	Ramp	Deck sheets were drawn self-intersecting and floated between the cheeks	Folded 3 mm trays the full inner width, flanges bolted through the cheeks; roller 10 mm proud of the deck ends	Fixings

Consequences

- `cad/src/model.py` holds the constructable design; STEP and STL, SLH-DWG-001 and 002 (Rev P2), the concept media and the build plan pictures are regenerated from it.
- SLH-CAL-001 is rerun on it; costs rise to USD 887.50, still USD 1,112.50 under the value-engineering target.
- `design_state: constructable` is set in `project.yaml`.